

CAPSULE-CARE

Smart Queue & Medicine Reminder System

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1. Abstract

Capsule-Care is a web-based smart queue and medicine reminder system for clinics and small hospitals. Patients book appointments, receive token numbers, and track queue position in real time. Doctors manage the queue, attend patients one by one, mark payments, and download daily reports. After consultation, patients get medicine dosage reminders. This eliminates long physical waiting lines and saves time for both patients and doctors.

2. Problem Statement

In most clinics, patients wait in long physical queues without knowing their turn. Doctors have no digital way to manage patient flow, track payments, or send medicine reminders. This leads to confusion, delays, and poor experience. There is no simple, affordable system combining queue management with medicine reminders.

3. Proposed Solution

Capsule-Care provides a token-based smart queue where patients book online, get a token, and wait from home. Doctors start the queue, attend patients, mark them done. Paid patients get priority. After consultation, patients receive medicine reminders. The system runs on any web browser with no app installation needed.

4. System Architecture

The system follows a 3-tier architecture with Flask backend, MySQL database, and HTML/CSS frontend.

Presentation Layer (Frontend)

- HTML5 + CSS3 + Jinja2 Templates
- Responsive UI with Dark/Light Theme
- Token Card PDF Download
- Real-time Queue Status Board

Application Layer (Backend)

- Python Flask Web Framework
- Session-based Authentication
- Queue Engine with Token Priority
- Payment Tracking (GPay/Cash)

Data Layer (Database)

MySQL on Aiven Cloud

Tables: patients, doctors, appointments,
reminders, queue_sessions

Auto-initialize on first run

5. Patient Flow



Register/Login: Patient creates account with name, age, email, mobile and password.

Book Appointment: Select doctor, date and time slot. System assigns a token number automatically.

Payment: GPay option for online payment or Cash paid at clinic. Paid patients get queue priority.

Queue Tracking: Live board shows all tokens with status (Waiting, Attending, Done, Postponed, Cancelled).

Token Card: Download PDF token with QR code, doctor name, date and token number.

Medicine Reminders: Set dosage, frequency, food timing and duration after consultation.

6. Doctor Flow



Register/Login: Doctor creates profile with name, specialization, fees and optional GPay QR.

Start Queue: Doctor clicks Start Queue button. Until then all patients see Waiting status.

Mark Paid/Unpaid: Doctor toggles payment status. Paid tokens jump ahead in queue priority.

Mark Done: Doctor marks token as completed. Patient token becomes inactive.

Postpone: Doctor can postpone whole day queue to a future date. Cancel Postpone restores waiting.

7. Key Features

Smart Queue Management -- Token-based queue with auto-numbering. Paid patients get priority. Real-time status board.

Postpone and Cancel -- Doctor postpones whole day queue. Patients see red alert with new date. Cancel Postpone restores waiting.

Payment Tracking -- GPay or Cash payment. Doctor toggles Paid/Unpaid. Paid tokens move ahead in queue.

Token PDF Download -- Downloadable appointment token card with QR code and details.

Medicine Reminders -- Dosage, frequency, food timing and duration with browser notifications.

Doctor Reports -- Daily PDF report of completed tokens with patient details.

Dark/Light Theme -- Toggle between themes saved in browser.

Responsive Design -- Works on mobile, tablet and desktop.

8. Database Schema

Table	Key Columns	Purpose
patients	id, name, age, email, mobile, password	Patient accounts
doctors	id, doctor_name, username, email, fees, gpay_qr	Doctor profiles
appointments	id, patient_id, doctor_id, token_number, status, payment_status, postponed_date	Bookings and queue
reminders	id, patient_id, medicine_name, dosage, food_timing	Medicine reminders
queue_sessions	doctor_id, queue_date, started_at	Queue start tracking

Statuses: waiting, done, postponed, absent, cancelled

Payment: pending, paid, cash

9. Technology Stack

Layer	Technology
Frontend	HTML5, CSS3, Jinja2 Templates, SVG Icons
Backend	Python 3.13, Flask 3.0.3
Database	MySQL 8.0 (Aiven Cloud)
PDF Generation	ReportLab 4.2.2
Deployment	Vercel Serverless / Render
Version Control	Git + GitHub
Auth	Session-based with Werkzeug hashing

10. Deployment Steps (Render)

1. Push source code to GitHub repository.
2. Go to render.com, sign up with GitHub.
3. Click New + > Web Service, select HACKATHON repo.
4. Set Runtime: Python, Build: pip install -r requirements.txt, Start: gunicorn app:app.
5. Add Environment Variables: DB_HOST, DB_PORT, DB_USER, DB_PASSWORD, DB_NAME, SECRET_KEY.
6. Click Create Web Service, wait 3-4 minutes.
7. Visit /setup endpoint once to create database tables.

11. Project Structure

app.py -- Main Flask application
schema.sql -- Database DDL
requirements.txt -- Python dependencies
Procfile -- Render start command
vercel.json -- Vercel config
api/index.py -- Serverless entry point
static/css/style.css -- Styles with dark/light theme
templates/ -- 18 Jinja2 HTML templates

12. Future Enhancements

- SMS/Email appointment confirmations
- Doctor availability calendar
- Patient review and rating
- Multi-language support (Tamil + English)
- Admin dashboard for hospital management

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